



IILM UNIVERSITY
INTERNSHIP REPORT
ON
C++ PROGRAMMING

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DECLARATION

I hereby declare that this internship report titled “C++ Programming Internship” is a record of the work carried out by me during my internship with CodeAlpha. The projects presented in this report, namely the Banking Management System and Sudoku Solver, were developed as part of the internship learning and project work. The report has been prepared for academic submission and reflects my understanding of the concepts and implementation work completed during the internship.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodeAlpha for providing me with the opportunity to work on a C++ programming internship. This internship gave me the opportunity to apply programming concepts to practical problems and improve my understanding of software development.

I am also thankful to my university faculty and mentors for their guidance and support. Finally, I would like to thank everyone who helped me directly or indirectly during the completion of the projects and this report.

ABSTRACT

This report presents the work completed during a C++ Programming Internship at CodeAlpha. The main objective of the internship was to strengthen practical programming skills by developing applications based on real-world requirements. Two projects were completed: a Banking Management System and a Sudoku Solver.

The Banking Management System is a console-based application developed using object-oriented programming and file handling. It provides functions such as account creation, deposits, withdrawals, fund transfers, account details, account closure, and transaction history. The Sudoku Solver uses a backtracking algorithm to solve a standard 9×9 Sudoku puzzle while checking row, column, and 3×3 sub-grid constraints.

Together, these projects provided practical experience in C++, classes and objects, functions, arrays, recursion, file handling, validation, algorithms, and problem solving.

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1. INTRODUCTION

C++ is a general-purpose programming language widely used for software development, system programming, competitive programming, game development, and applications where performance and control over system resources are important. Learning C++ also provides a strong foundation for understanding data structures, algorithms, object-oriented programming, and problem solving.

The internship provided an opportunity to move from classroom-based programming to practical project development. Instead of focusing only on individual programming questions, the projects required planning features, writing reusable functions, validating input, handling files, and testing the complete application.

2. INTERNSHIP OBJECTIVES

- Strengthen the fundamentals of C++ programming.
- Understand and apply object-oriented programming concepts.
- Develop programs that solve practical problems.
- Improve logical thinking and algorithm design.
- Learn file handling for persistent data storage.
- Understand recursion and backtracking through a practical problem.
- Practice input validation and error handling.
- Improve debugging, testing, and code organization skills.

3. ORGANIZATION AND INTERNSHIP OVERVIEW

The internship was completed under CodeAlpha's C++ Programming Internship program. The internship focused on practical C++ programming and software development. The assigned work involved completing programming projects and applying core programming concepts to functional applications.

As part of the internship work, the two selected projects were a Banking Management System and a Sudoku Solver. These projects cover different areas of programming: the banking system focuses mainly on object-oriented design, data management and file handling, while the Sudoku Solver focuses on algorithms, recursion and backtracking.

4. PROJECT 1 – BANKING MANAGEMENT SYSTEM

4.1 Project Overview

The Banking Management System is a console-based C++ application designed to simulate basic banking operations. The purpose of the project is to provide a simple way to create and manage customer accounts while maintaining account balances and transaction records.

4.2 Objectives

- Create and manage bank accounts.
- Allow customers to deposit and withdraw money.
- Transfer money between accounts.
- Display account details and current balance.
- Maintain transaction history.
- Store account information using files so that data can be retained between program runs.
- Validate user input and prevent invalid banking operations.

4.3 Main Features

Account Creation: Creates a new account and assigns an account number.

Deposit: Adds a valid amount to an existing account balance.

Withdrawal: Withdraws money after checking the available balance.

Fund Transfer: Transfers an amount from one account to another after validation.

Account Details: Displays relevant account information and balance.

Close Account: Allows account closure when the required conditions are satisfied.

Display Accounts: Shows active accounts maintained by the system.

Transaction History: Records important account transactions in a separate file.

File Handling: Stores account information in accounts.txt and transactions in transactions.txt.

4.4 Technologies and Concepts

- C++ programming language
- Classes and objects
- Functions
- Conditional statements and loops
- File input/output
- Vectors/arrays and data storage
- Input validation
- Basic exception/error handling

4.5 Working of the System

When the application starts, previously saved account information can be loaded from the account data file. The user is presented with a menu containing the available operations. For an account-related operation, the program identifies the required account and performs the requested action after checking the input.

Deposits increase the account balance, while withdrawals and transfers reduce the balance only when sufficient funds are available. Each important operation can be recorded in the transaction history file. The updated information is saved so that it can be available when the program is opened again.

4.6 Input Validation

Input validation is used to reduce incorrect operations. Examples include checking that amounts are positive, verifying that an account exists, preventing withdrawals beyond the available balance, and ensuring that account closure follows the required conditions.

4.7 Project Outcome

The completed system provides the core functionality expected from a small console-based banking application. The project demonstrated how C++ classes, functions, file handling and validation can be combined to build a complete application rather than isolated programs.

5. PROJECT 2 – SUDOKU SOLVER

5.1 Project Overview

The Sudoku Solver is a C++ program that solves a standard 9×9 Sudoku puzzle. Empty cells are represented by 0. The program searches for valid values and uses the backtracking algorithm to find a complete solution.

5.2 Objectives

- Represent a Sudoku puzzle using a two-dimensional array.
- Identify empty cells in the puzzle.
- Check whether a number can be placed safely.
- Use recursion to explore possible solutions.
- Apply backtracking when a selected number leads to an invalid state.
- Display the solved puzzle or report that no solution exists.

5.3 Algorithm Used – Backtracking

Backtracking is a recursive problem-solving technique. The solver selects an empty cell and tries possible numbers from 1 to 9. Before placing a number, it checks whether the number already exists in the same row, column, or 3×3 sub-grid.

If a number is valid, it is placed in the cell and the solver recursively moves to the next empty cell. If the later steps cannot produce a solution, the program removes the previously placed number and tries another possibility. This process continues until the puzzle is solved or all possibilities have been exhausted.

5.4 Sudoku Constraints

- Each row must contain the numbers 1 to 9 without repetition.

- Each column must contain the numbers 1 to 9 without repetition.
- Each 3×3 sub-grid must contain the numbers 1 to 9 without repetition.

5.5 Main Functions

findEmptyCell(): Finds an empty position in the Sudoku grid.

isSafe(): Checks row, column and 3×3 sub-grid constraints before placing a number.

solveSudoku(): Uses recursion and backtracking to solve the puzzle.

printGrid(): Displays the Sudoku grid in a readable format.

5.6 Project Outcome

The Sudoku Solver successfully demonstrates the use of recursion and backtracking. It provides a practical example of how an algorithm can systematically explore possible solutions and return to an earlier decision when that decision does not lead to a valid result.

6. TECHNOLOGIES AND CONCEPTS USED

Concept / Technology	Application in Projects
C++	Primary programming language used for both projects.
Object-Oriented Programming	Used mainly in the Banking Management System for organizing account-related data and operations.
Classes and Objects	Used to model banking accounts and their operations.
Functions	Used to divide the programs into smaller reusable tasks.
File Handling	Used to store account and transaction information.
Arrays / Data Structures	Used for representing and managing Sudoku data.
Recursion	Used in the Sudoku solving process.
Backtracking	Used to search for a valid Sudoku solution.
Input Validation	Used to prevent invalid values and operations.
Debugging and Testing	Used to identify and correct logical and runtime issues.

7. TESTING AND RESULTS

Testing was performed by providing normal inputs as well as invalid inputs to verify that the applications behaved as expected.

Project	Test Case	Expected Result
Banking System	Create a new account	Account is created with a

		unique account number.
Banking System	Deposit a valid amount	Balance increases by the deposited amount.
Banking System	Withdraw more than balance	Transaction is rejected.
Banking System	Transfer between valid accounts	Amount is deducted and credited correctly.
Banking System	Close account with outstanding balance	Account closure is rejected.
Sudoku Solver	Provide a valid solvable puzzle	Program displays the solved Sudoku.
Sudoku Solver	Provide an unsolvable puzzle	Program reports that no solution is available.
Sudoku Solver	Check invalid placement	Number is not placed when it violates Sudoku rules.

8. CHALLENGES FACED AND SOLUTIONS

Managing account data: Structured the banking application using classes and organized functions to make account operations easier to manage.

Persistent storage: Used file handling so account and transaction information can be saved and loaded.

Preventing invalid banking operations: Added checks for account existence, positive amounts, available balance and other conditions.

Understanding recursion: Broke the Sudoku solving process into small recursive steps and used a safe-placement check.

Backtracking logic: Removed an earlier choice whenever it prevented the puzzle from reaching a valid solution.

Debugging: Tested individual operations and corrected logical errors through repeated execution.

9. LEARNING OUTCOMES

- Improved understanding of C++ syntax and program structure.
- Gained practical experience with object-oriented programming.
- Learned how to design a menu-driven console application.
- Improved understanding of file input/output.
- Learned how recursion works in a practical algorithm.
- Understood the backtracking approach for constraint-based problems.
- Improved debugging and testing skills.
- Learned how to organize programming projects for GitHub submission.
- Gained confidence in converting a problem statement into a working program.

10. CONCLUSION

The C++ Programming Internship provided valuable practical experience in software development and problem solving. The Banking Management System and Sudoku Solver were useful projects for applying different C++ concepts in realistic programming situations.

The banking project strengthened understanding of object-oriented programming, file handling, validation and application design. The Sudoku project strengthened understanding of recursion, backtracking and algorithmic thinking. Overall, the internship helped bridge the gap between theoretical programming concepts and their practical implementation.

11. FUTURE SCOPE

- Add a graphical user interface to both applications.
- Use a database instead of text files for the banking system.
- Add secure authentication and role-based access to the banking system.
- Add more detailed transaction reports and search/filter options.
- Allow users to enter Sudoku puzzles through a graphical interface.
- Add Sudoku difficulty levels and puzzle generation.
- Improve project documentation and automated testing.

12. REFERENCES

- CodeAlpha – C++ Programming Internship Tasks and Instructions.
- C++ language documentation and standard programming references.
- Standard concepts of recursion and backtracking algorithms.
- Project source code developed during the internship.

END OF REPORT

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